

1. Run `task_0a.py` to generate the vector database if not already generated.
2. Press Run All (or restart kernel and run all cells).
3. You will be prompted to provide input values.

```
In [ ]: LABEL = input("Provide a valid label name / label ID present in the Caltech101 dataset\n")  
  
        FEATURE_SPACE = input("Provide a feature space [color, hog, avgpool, layer3, layer4]\n")  
  
        K = int(input("Enter K, the number of similar images K to find for the selected label\n"))
```

```
Input label: 20
Downloading dataset if not present.
Files already downloaded and verified
Output label: ceiling_fan
```

```
In [ ]: # Load feature space and representative label vectors for that feature space

from utils.database_utils import retrieve

feature_vectors = retrieve(f'{FEATURE_SPACE}.pt')
rep_label_vectors = retrieve(f'rep_label_{FEATURE_SPACE}.pt')
```

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```
for i in range(len(top_k)):

    distance, image_id, label = top_k[i]

    print(str(i + 1) + ".", "\tImage ID: ", image_id, "\t\tDistance: ", distance)
    display(dataset[image_id][0])
```

Downloading dataset if not present.
Files already downloaded and verified

K = 10 most similar images by fc feature space using distance: cosine

1. Image ID: 3648 Distance: 0



2. Image ID: 3688 Distance: 0



3. Image ID: 3684 Distance: 0.08956003189086914



4. Image ID: 3652

Distance: 0.101817786693573



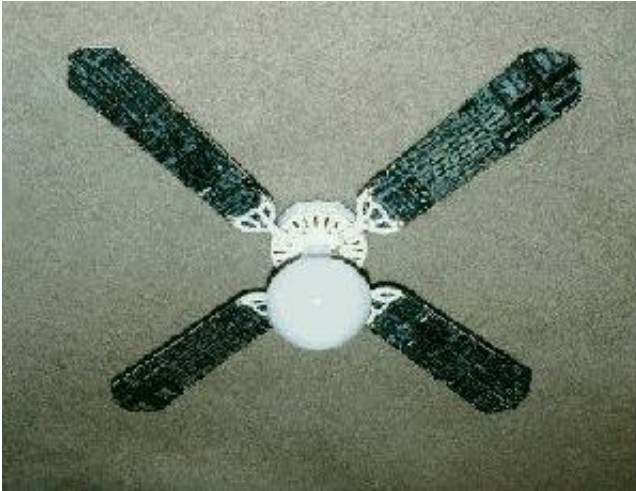
5. Image ID: 3694

Distance: 0.15001046657562256



6. Image ID: 3678

Distance: 0.15082710981369019



7. Image ID: 3656

Distance: 0.1555192470550537



8. Image ID: 3664

Distance: 0.15681815147399902



9. Image ID: 3676

Distance: 0.16604405641555786



10. Image ID: 3666

Distance: 0.16704869270324707

